Circuit Diagrams, Design Values and Component Roles

A. Two-stage CMOS op-amp (the analog engine) – Cadence Virtuoso

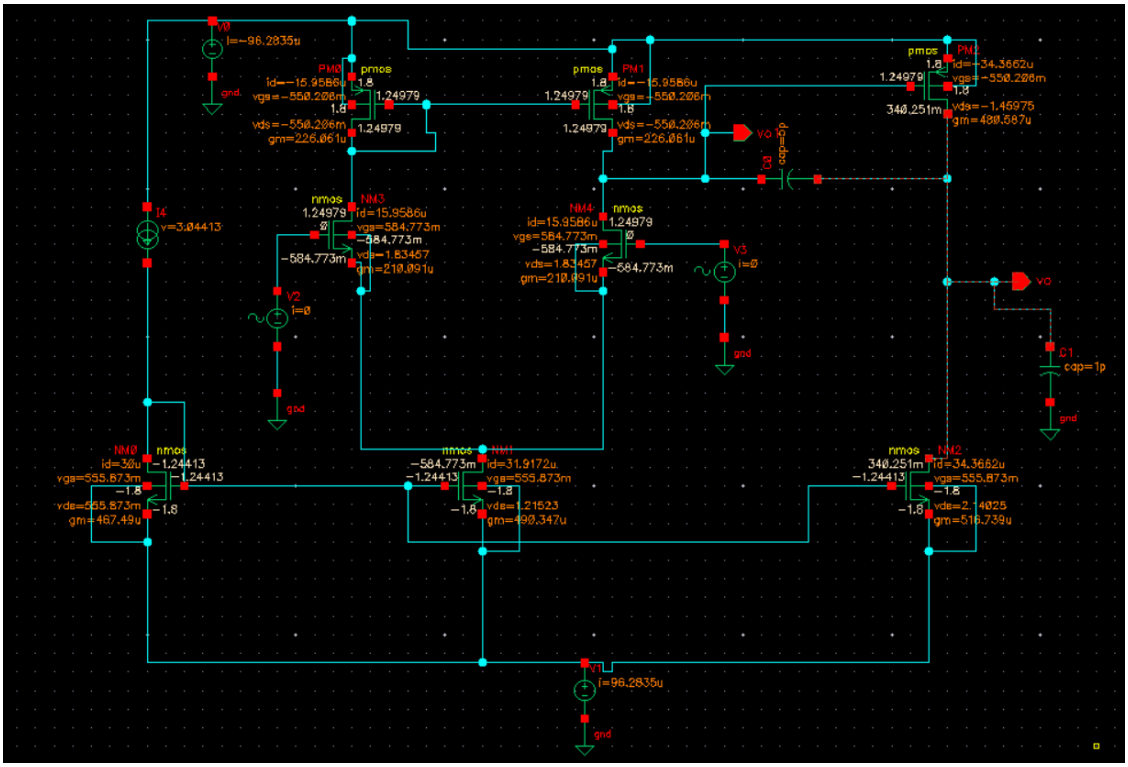


Figure A1. Op-amp schematic in Cadence Virtuoso ADE with simulated DC operating points (id, vgs, vds, gm) annotated on every transistor.

The op-amp is a two-stage, Miller-compensated OTA: an NMOS-input differential pair with a PMOS current-mirror load (Stage 1) drives a PMOS common-source stage biased by an NMOS current source (Stage 2). It runs from a single **1.8 V** supply, every device has **L = 500 nm**, and the load is **C1 = 1 pF**. Sizing uses the gm/Id method instead of the square-law V_ov approximation, because the gm/Id curves are extracted from simulated device data and stay valid in weak, moderate and strong inversion.

Design target: 5000 V/V ≤ A_v,dc ≤ 6000 V/V. **Achieved (simulated):** 5643.35 V/V (75.03 dB).

Device list and why these values

Device	Type, W/L (µm)	Role	Why this value
NM0 / NM1 / NM2	NMOS, L = 0.5	Bias mirror: reference, Stage-1 tail source, Stage-2 current source	Same gate bias. NM0 carries the 30 µA reference, NM1 the 31.917 µA tail (2 × 15.959 µA), NM2 the 34.366 µA Stage-2 current.
NM3, NM4	NMOS 3 / 0.5	Input differential pair	gm/Id = 13.25 V^{-1} and Id/W = 5.30 µA/µm at 15.959 µA each gives W ≈ 3 µm, gm ≈ 211 µS.
PM0, PM1	PMOS 20 / 0.5	Current-mirror load	gm/Id = 13.90 V^{-1} and Id/W = 0.7975 µA/µm at 15.959 µA gives W ≈ 20 µm.
PM2	PMOS 40 / 0.5	Second-stage common-source amplifier	gm/Id = 13.71 V^{-1} and Id/W = 0.8591 µA/µm at 34.366 µA gives W ≈ 40 µm, gm ≈ 471 µS.
NM2 (as load of PM2)	NMOS ≈ 10 / 0.5	Second-stage current source	Carries 34.366 µA (matches PM2 within 0.006 %). Its r_out of 374.26 kΩ is taken directly from the simulated operating point.
C0 (Cc)	5 pF	Miller compensation	Connects the Stage-1 output to the op-amp output and sets the dominant pole, giving PM ≈ 56.7°.
C1 (CL)	1 pF	Load capacitor	Output load for all AC and transient tests.
VDD	1.8 V	Supply	Single supply. Total current from the three bias branches ≈ 96.28 µA (≈ 173 µW).

How W, gm and ro are obtained from the gm/Id curves

For each device type, a fixed-L (500 nm) test transistor is DC-swept and Id/W, gm/Id and gm/gds are plotted against V_GS (or V_SG). The bias point is read at the marker, and the three quantities below follow from the branch current Id taken from the sized schematic:

W = Id / (Id/W) *gm = (gm/Id) × Id* *ro = 1/gds = (gm/gds) / gm*

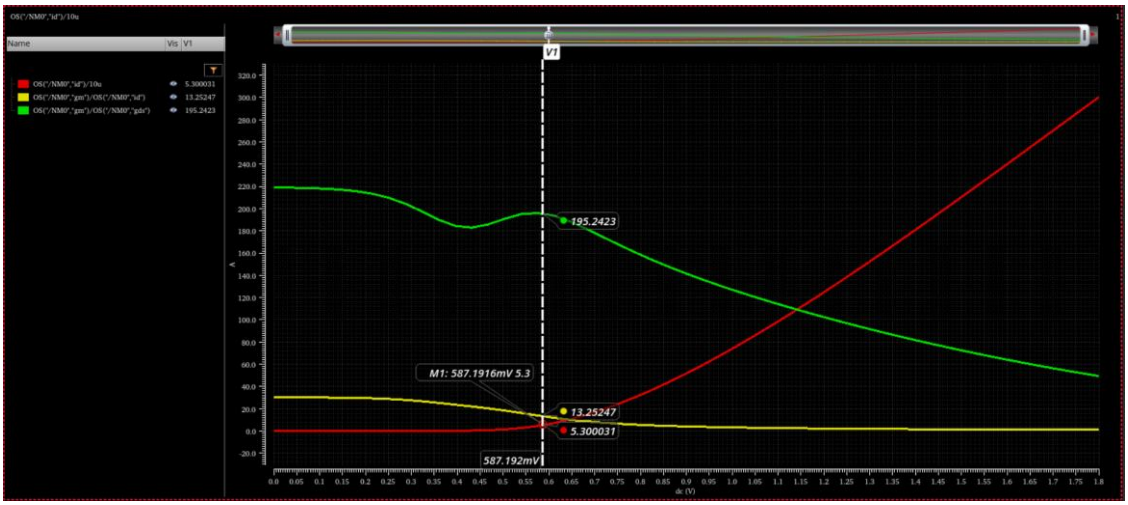


Figure A2. Id/W, gm/Id and gm/gds vs. V_GS for the Stage-1 NMOS (marker at V_GS = 587.19 mV).

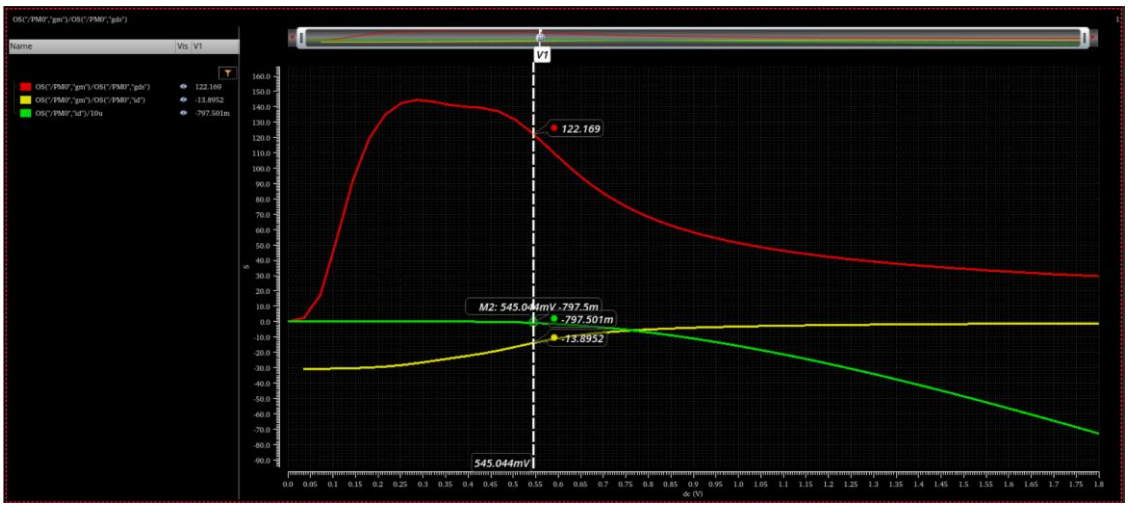


Figure A3. Id/W, gm/Id and gm/gds vs. V_SG for the Stage-1 PMOS mirror device (marker at V_SG = 545.04 mV).

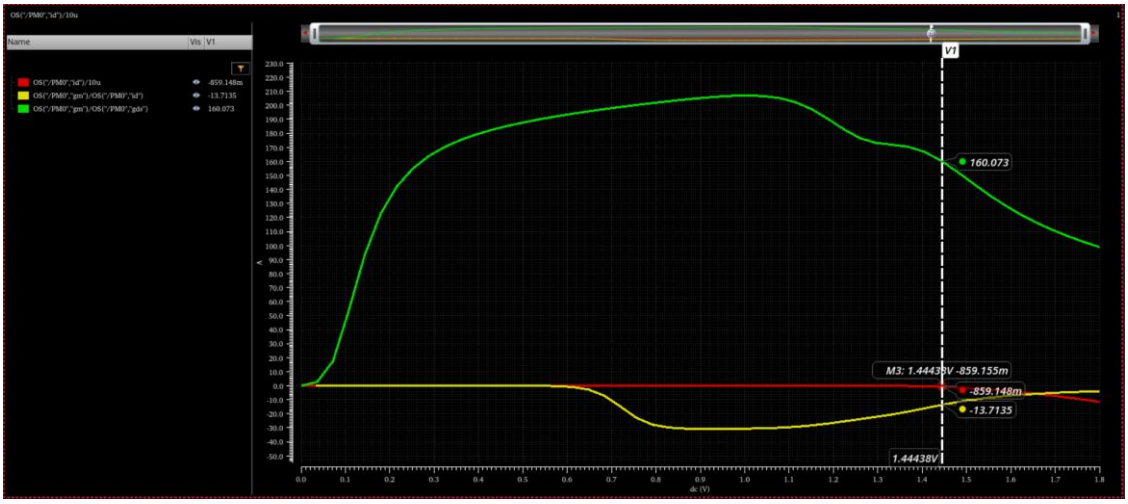


Figure A4. Id/W, gm/Id and gm/gds vs. V_SG for the Stage-2 PMOS common-source device (marker at V_SG = 1.44438 V).

Device	gm/Id (V^{-1})	gm/gds	Id/W (µA/µm)	Id (µA)	W = Id/(Id/W)	gm (µS)	ro (kΩ)
NM3 / NM4	13.2525	195.242	5.3000	15.9586	3.011 µm	211.49	923.17
PM0 / PM1	13.8952	122.169	0.7975	15.9586	20.011 µm	221.75	550.94
PM2	13.7135	160.073	0.8591	34.3662	40.00 µm	471.28	339.66
NM2	15.036	—	—	34.3662	≈ 10 µm	516.73	374.26

> NM2 is not re-derived from a characterization sweep: its r_out is the simulated small-signal value, and its gm/Id, W and gm are as listed in the Cadence report summary table.

Expected performance (hand calculation vs. Spectre)

Item	Value
Stage 1 output resistance	$R_{out1} = r_{o,n} \parallel r_{o,p} = 923.17\ k\Omega \parallel 550.94\ k\Omega = 345.03\ k\Omega$
Stage 1 gain (diff pair)	$A_{v1} = g_{m,n} \cdot R_{out1} = 0.2115\ mS \times 345.03\ k\Omega \approx 72.97\ V/V$ (37.3 dB)
Stage 2 output resistance	$R_{out2} = r_{o,p2} \parallel R_{out,n2} = 339.66\ k\Omega \parallel 374.256\ k\Omega = 178.06\ k\Omega$
Stage 2 gain	$A_{v2} = g_{m,p2} \cdot R_{out2} = 0.4713\ mS \times 178.06\ k\Omega \approx 83.92\ V/V$ (38.5 dB)
Total DC gain, hand calculation	$A_{v1} \cdot A_{v2} \approx 6123\ V/V$ (75.74 dB)
Total DC gain, Spectre	5643.35 V/V (75.03 dB), within the 5000–6000 V/V target; hand value is 8.5 % higher
Supply current / power	≈ 96.28 µA / ≈ 173 µW at 1.8 V

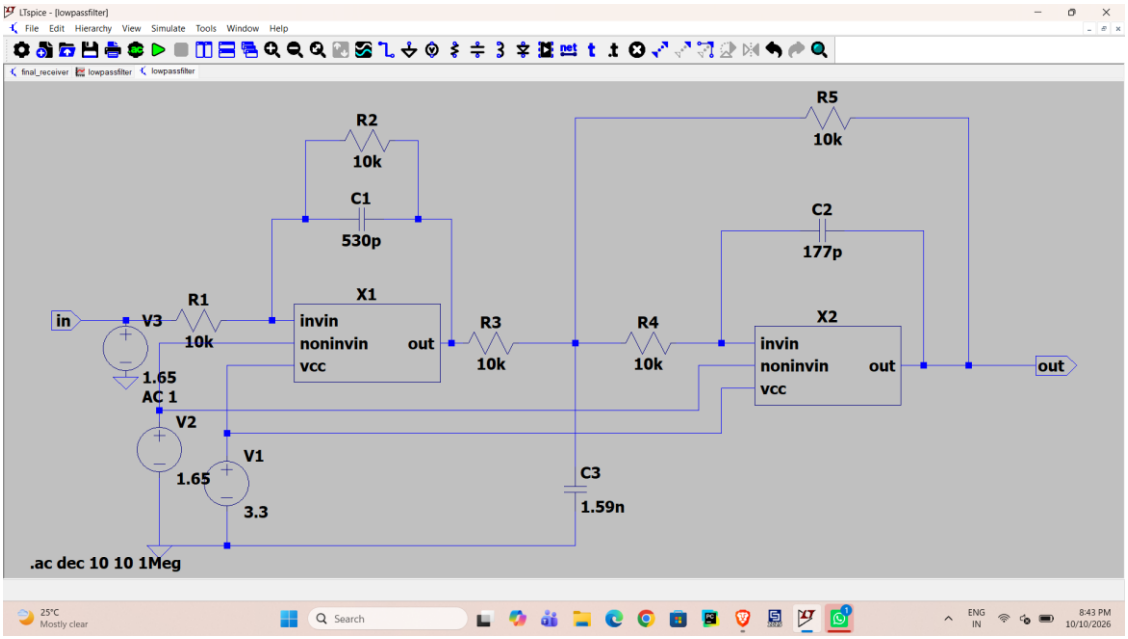
The hand estimate is higher because it ignores channel-length modulation in the bias network, body effect and mismatch. As a second check, the hand-calculated gm values are within 2 % of the gm annotated on the schematic (NM3 211.49 vs. 210.09 µS, PM0 221.75 vs. 226.06 µS, PM2 471.28 vs. 480.59 µS).

Compensation and dynamic performance (Spectre)

Metric	Value
Open-loop DC gain	5643.35 V/V (75.03 dB)
−3 dB bandwidth	1.167 kHz
Gain-bandwidth product ($A_{v,dc} \times f_{-3dB}$)	≈ 6.59 MHz
Unity loop-gain frequency	7.33 MHz
Phase margin	≈ 56.7° (inside the 45°–60° guideline)
Slew rate	≈ 6.66 V/µs

> Scope of this change: only the op-amp (section A) was Cadence design. The filter and receiver in sections B and C, with their 3.3 V supply and 1.65 V reference, are the LTspice circuits as simulated.

B. 3rd-order Butterworth active-RC low-pass filter



Low-pass filter schematic

The filter is two op-amp stages: a first-order inverting section (X1) followed by a multiple-feedback (MFB) biquad (X2). Both inverting stages cancel, so the DC gain is +1. The non-inverting inputs are tied to V_ref = 1.65 V (V2).

Transfer function

A 3rd-order Butterworth filter has one real pole and a complex pair with Q = 1:

$$H(s) = \frac{1}{1 + s/\omega_c} \cdot \frac{\omega_c^2}{s^2 + \omega_c s + \omega_c^2}, \qquad \omega_c = 2\pi \cdot 30\ \text{kHz}$$

Component values

Part	Value	Role
R1	10 kΩ	Input resistor of section 1 (sets gain −R2/R1 = −1)
R2	10 kΩ	Feedback resistor of section 1
C1	530 pF	Across R2: pole at $1/(2\pi R_2 C_1)$ = 30.0 kHz
R3	10 kΩ	Input resistor of the biquad
R5	10 kΩ	Feedback resistor from the biquad node to the output
R4	10 kΩ	Resistor from the biquad node to the inverting input
C3	1.59 nF	Biquad node to ground: sets the pole frequency
C2	177 pF	Op-amp feedback capacitor: sets the damping (Q)
V2	1.65 V	Reference for the non-inverting inputs (mid-supply)
V1	3.3 V	Supply

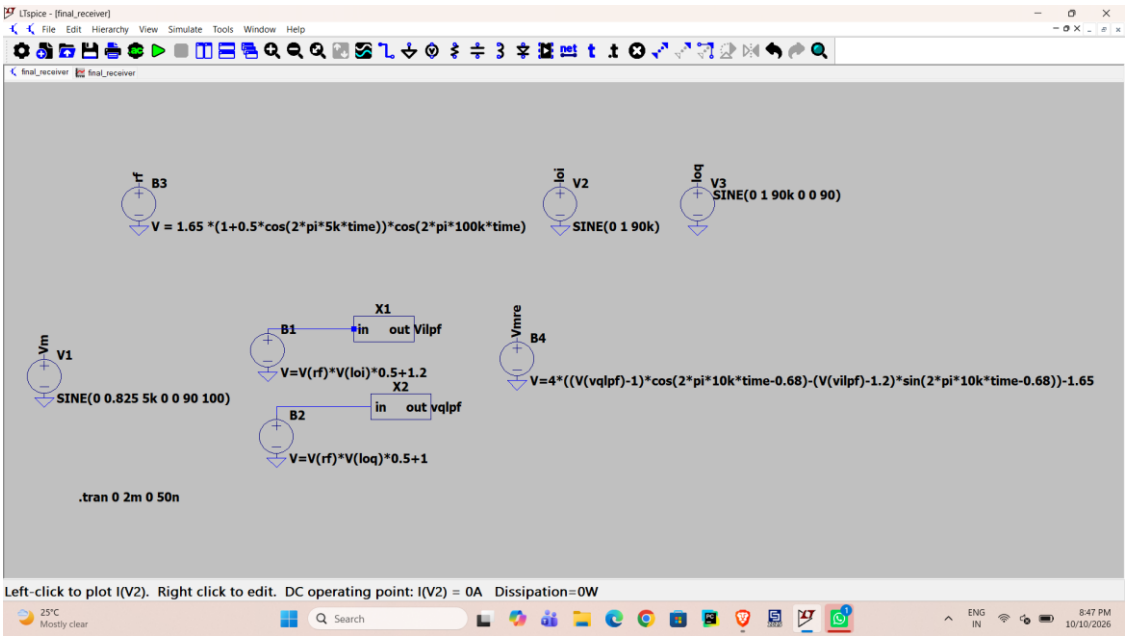
Why these values

For the MFB section with all resistors equal to R and unity gain:

$$\omega_0 = \frac{1}{R\sqrt{C_2 C_3}}, \qquad Q = \frac{1}{3}\sqrt{\frac{C_3}{C_2}}$$

- Q = 1 requires $C_3 = 9\,C_2$.
- With R = 10 kΩ and $\omega_0 = 2\pi \cdot 30$ kHz: $C_2 = 1/(3R\,\omega_0) = 177$ pF and $C_3 = 9\,C_2 = 1.59$ nF.
- First section: $C_1 = 1/(2\pi R \cdot 30\text{ kHz}) = 530$ pF.
- **Why 30 kHz:** the IF is 10 kHz and the message sidebands extend to 15 kHz. A 30 kHz cutoff passes both with about 0.1 dB loss (gain 0.988 at 15 kHz) and rejects the 190 kHz mixer sum product by 48 dB. A 100 kHz cutoff left too much of the sum term, and mismatched capacitors (a cutoff near 11 kHz) attenuated the IF itself.
- **Why 10 kΩ:** low enough for modest thermal noise and small capacitors, high enough that the op-amp can drive it comfortably.
- **Why a Butterworth response:** maximally flat passband, so the IF and its sidebands are not distorted.

C. Complete receiver (top level)



Receiver schematic

Block	Expression / value	Purpose
V1 (vm)	`SINE(0 0.825 5k 0 0 90 100)`	Reference message, 0.825 cos(2π·5k·t)
B3 (rf)	`1.65*(1+0.5*cos(2*pi*5k*time))*cos(2*pi*100k*time)`	AM RF signal: carrier 1.65 V, μ = 0.5
V2 (loi)	`SINE(0 1 90k)`	In-phase LO: sin(2π·90k·t)
V3 (loq)	`SINE(0 1 90k 0 0 90)`	Quadrature LO: cos(2π·90k·t)
B1	`V(rf)*V(loi)*0.5+1.2`	I mixer: product × 0.5, plus 1.2 V offset
B2	`V(rf)*V(loq)*0.5+1`	Q mixer: product × 0.5, plus 1.0 V offset

Block	Expression / value	Purpose
X1, X2	`lowpassfilter` (section B)	3rd-order Butterworth, 30 kHz, one per path
B4 (vmre)	`4*((V(vqlpf)-1)*cos(2*pi*10k*time-0.68)-(V(vilpf)-1.2)*sin(2*pi*10k*time-0.68))-1.65`	Second quadrature mixer, scaling and DC removal
B5	`sqrt(V(Vm)^2)`	Unused test source
.tran	`0 2m 0 50n`	2 ms simulation, 50 ns maximum step

Why these values

- **LO 90 kHz, IF 10 kHz.** An IF different from the message frequency avoids confusing the IF beat with the message (with an IF equal to 5 kHz the "recovered" tone was the IF itself, not the message). 10 kHz also keeps the 5 kHz sidebands (5 to 15 kHz) clear of DC.
- **Mixer scale 0.5 and offsets 1.2 V / 1.0 V.** The scale keeps the raw product (up to ±1.24 V) inside the op-amp range. The offsets center each path in the 0 to 3.3 V window. The first filter stage output sits at 3.3 − V_offset (2.1 V for I, 2.3 V for Q) and swings about ±0.6 V, so the Q path is the closer of the two to the upper limit of the op-amp. Using 1.2 V for both paths would give the Q path more headroom (then replace −1 by −1.2 in B4).
- **Phase 0.68 rad.** Equals the filter phase lag at 10 kHz (39°). Without it the output is scaled by cos 39° = 0.78.
- **Gain 4 and −1.65 V.** The factor 4 cancels the mixer gain (0.5) and the 1/2 from the product-to-sum identity, giving A_c + m(t); subtracting A_c = 1.65 V leaves m(t).
- **Reference message.** V(vm) is plotted only to compare with V(vmre).

D. Component significance, summary

Component	Significance
Differential pair (NM3, NM4)	Amplifies the voltage difference; sets gm and so the gain-bandwidth
Current mirror (PM0, PM1)	Converts the differential current to a single-ended output and doubles the gain
Tail source (NM1)	Sets total input-stage current and rejects common-mode signals
Common-source stage (PM2, NM2)	Provides the second gain stage and a larger output swing
Miller capacitor (Cc)	Dominant pole: makes the amplifier stable in feedback (PM ≈ 56.7°)
gm/Id sizing	Sets W, gm and ro of each device from simulated characterization curves
Filter capacitors C1, C2, C3	Set the pole frequencies and the damping of the Butterworth response
Filter resistors (10 kΩ)	Set the gains (unity) and, with the capacitors, the pole frequencies
Reference 1.65 V	Biases the op-amps at mid-supply so signals can swing both ways on one supply
I/Q mixers	Translate the RF band down to the IF in two quadrature paths
Second quadrature mixer	Combines I and Q to remove the IF carrier and reconstruct the message